

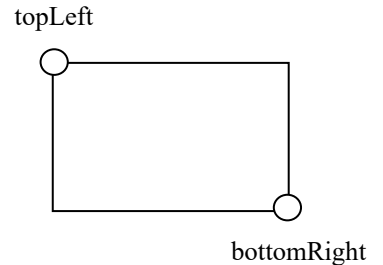
rectangle

A structure called Point is defined to represent a point in 2D which is given as follows:

```
typedef struct {
    double x;
    double y;
} Point;
```

Another structure called Rectangle is defined as follows:

```
typedef struct {
    Point topLeft; /* top left point of rectangle */
    Point botRight; /* bottom right point of rectangle */
} Rectangle;
```



Write a C program that reads in the top left point and bottom right point of a rectangle, computes the area of the rectangle and prints the area of the rectangle on the screen. Your program should include the following three functions with prototypes:

- (1) `void getRect(Rectangle *r);` /* read in the two points of rectangle */
- (2) `void printRect(Rectangle r);` /* print the coordinates of two points of rectangle */
- (3) `double findArea(Rectangle r);` /* return the area of rectangle */

A sample program template is given below to test the functions:

```
#include <stdio.h>
#include <math.h>
typedef struct {
    double x;
    double y;
} Point;
typedef struct {
    Point topLeft; /* top left point of rectangle */
    Point botRight; /* bottom right point of rectangle */
} Rectangle;
void getRect(Rectangle *r);
void printRect(Rectangle r);
double findArea(Rectangle r);
int main()
{
    Rectangle r;
    int choice;

    printf("Select one of the following options:\n");
    printf("1: getRect()\n");
    printf("2: findArea()\n");
    printf("3: printRect()\n");
    printf("4: exit()\n");
    do {
        printf("Enter your choice: \n");
```

```

scanf("%d", &choice);
switch (choice) {
    case 1:
        printf("getRect(): \n");
        getRect(&r);
        break;
    case 2:
        printf("findArea(): %.2f\n", findArea(r));
        break;
    case 3:
        printf("printRect(): \n");
        printRect(r);
        break;
    default:
        break;
}
} while (choice < 4);
return 0;
}
void getRect(Rectangle *r)
{
    /* Write your code here */
}
void printRect(Rectangle r)
{
    printf("Top left point: %.2f %.2f\n", r.topLeft.x, r.topLeft.y);
    printf("Bottom right point: %.2f %.2f\n", r.botRight.x, r.botRight.y);
}
double findArea(Rectangle r)
{
    /* Write your code here */
}

```

Some test input and output sessions are given below:

(1) Test Case 1:

Select one of the following options:

- 1: getRect()
- 2: findArea()
- 3: printRect()
- 4: exit()

Enter your choice:

1

getRect():

Enter top left point:

1 2

Enter bottom right point:

2 1

Enter your choice:

3

printRect():

Top left point: 1.00 2.00
Bottom right point: 2.00 1.00
Enter your choice:
4

(2) Test Case 2:
Select one of the following options:
1: getRect()
2: findArea()
3: printRect()
4: exit()
Enter your choice:
1
getRect():
Enter top left point:
1 2
Enter bottom right point:
2 1
Enter your choice:
2
findArea(): 1.00
Enter your choice:
4

(3) Test Case 3:
Select one of the following options:
1: getRect()
2: findArea()
3: printRect()
4: exit()
Enter your choice:
1
getRect():
Enter top left point:
1 5
Enter bottom right point:
5 5
Enter your choice:
2
findArea(): 0.00
Enter your choice:
4

(4) Test Case 4:
Select one of the following options:
1: getRect()
2: findArea()
3: printRect()
4: exit()
Enter your choice:
1

```
getRect():  
Enter top left point:  
1 5  
Enter bottom right point:  
2 2  
Enter your choice:  
3  
printRect():  
Top left point: 1.00 5.00  
Bottom right point: 2.00 2.00  
Enter your choice:  
2  
findArea(): 3.00  
Enter your choice:  
4
```